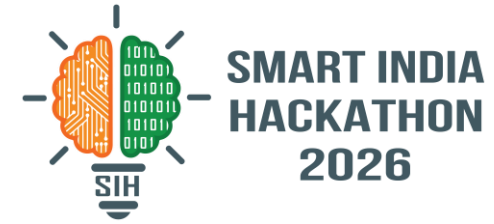
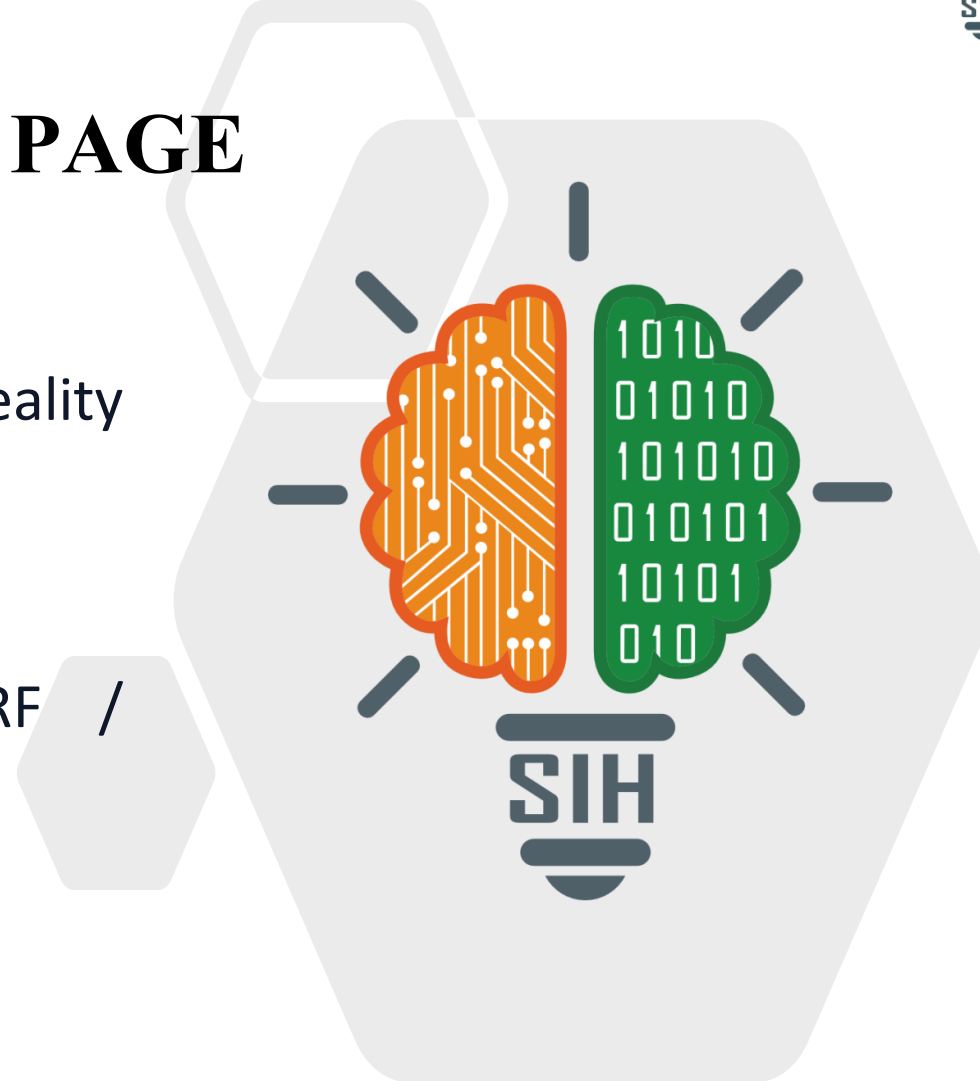


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** –SIH260088
- **Problem Statement Title-** Virtual Reality
& WebGL CBRN Emergency Simulator
- **Theme-**Disaster Management (NDRF /
MHA)
- **PS Category-** Software
- **Team ID-**



CBRS-X : ZERO-RISK CBRN HAZMAT SIMULATION & AUDIT ECOSYSTEM

PROPOSED SOLUTION & INNOVATION

1. THE PROBLEM ADDRESSED



Lethal Hazard Risks:

Live CBRN drills risk deadly chemical vapor and radiation exposure.



Prohibitive Consumable Cost:

Single-use Level-A hazmat suits cost ₹50,000+ per run.



Lack of Muscle Memory:

Classrooms fail to build real instinctive emergency reflex.



Subjective Grading Bias:

Human observation misses millisecond protocol errors.



2. PROPOSED SOLUTION



Zero-Risk 3D Simulation:

High-fidelity digital twin of Industrial Storage Bay 03 hot zone environment.



Zero-Install WebGL Access:

Runs natively in Chrome/Edge on standard office laptops.



Official 5-Stage SOP:

Enforces Detect → Protect → Contain → Evacuate → Decon.



Deterministic Scoring:

100-point audit engine checks safety breaches objectively.

3. INNOVATION & NOVELTY



Dual-Screen Ecosystem:

Real-time sync between Trainee WebGL Station & Instructor Radar HUD.



Sub-50ms Telemetry:

STOMP WebSockets stream responder data at 60Hz.



SHA-256 Certification:

Issues tamper-proof PDF credentials upon passing.



Polymorphic Hazards:

Supports Chemical, Bio & Radiological disaster runs.

5-STAGE TACTICAL LIFECYCLE

1. DETECT



PID Gas Sensor
PPM Concentration
Wind Direction

2. PROTECT



Level-A Hazmat Suit
SCBA Positive-Pressure
Donning

3. CONTAIN



Magnetic Patch
Application
Torque Clamp Seal

4. EVACUATE



Casualty Extraction
Warm Zone Triage

5. DECONTAMINATE

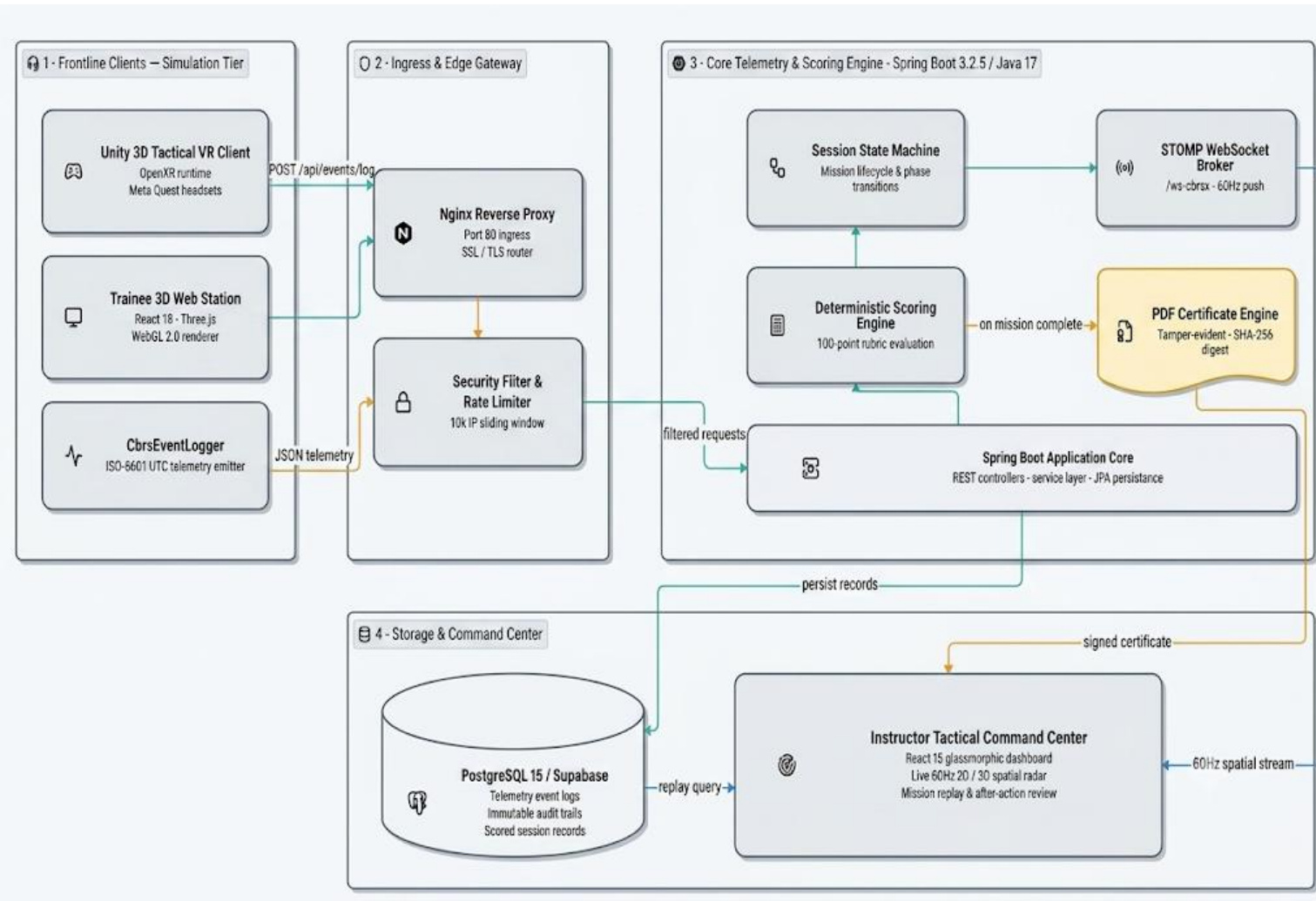


Multi-Stage Washdown
Archway
Dwell Time Check



Your
Team
Name

TECHNICAL APPROACH



TECHNOLOGY STACK

Unity + WebGL – VR simulation environment

React + Three.js – Trainee monitoring interface

Java + Spring Boot – Backend and real-time communication

PostgreSQL – Secure storage of trainee data and results

HOW IT WORKS

Simulate → Capture Actions → Evaluate Performance → Live Monitoring → Generate Certificate

Tracks trainee actions in **real time**

Automatically calculates a **100-point performance score**

Instructor can monitor trainees through a **live tactical view**

Generates a **secure, tamper-proof certificate**

PERFORMANCE

✓ 46/46 Tests Passed

✓ Supports up to 10,000 IP requests

✓ Smooth 60 FPS VR/WebGL experience

Your
Team
Name

FEASIBILITY AND VIABILITY



1. TECHNICAL & OPERATIONAL FEASIBILITY



Easy to Run

Works on standard laptops using Chrome/Edge.



Quick Deployment

Full system can be launched using Docker.



Supports Multiple Trainees

Handles multiple simulation sessions simultaneously.



Modular Scenarios

Easily supports Chemical, Biological & Radiological training scenarios.



2. RISK ANALYSIS & MITIGATION STRATEGY

Challenge

Our Solution



Heavy 3D loading



Compression & caching reduce load and enable faster launches.



No internet in field



Offline "Field Box" setup runs the entire system locally.



Fake certificates



SHA-256 verification ensures tamper-proof certificates.



Cybersecurity threats



Rate limiting & input security protect against attacks.



VERIFIED PERFORMANCE



46/46 Tests Passed



<50ms Telemetry



60 FPS Performance



100% RISK ELIMINATION

Zero physical injury, chemical burns or toxic vapor inhalation during emergency training drills.



90%+ COST REDUCTION

Saves ₹50,000 – ₹1,00,000 per responder per drill by eliminating suit waste and chemical reagents.



1,000+ DRILLS / DAY

Infinite virtual repetitions with automated deterministic grading and instant replay debriefs.



TARGET AUDIENCE & INSTITUTIONAL REACH



National & State Disaster Response Forces:

NDRF battalions, SDRF units across all states, and Civil Defense brigades.



Petrochemical & Heavy Industrial Facilities:

Oil refineries, chemical depots, and nuclear power plants (IOCL, ONGC, GAIL, NTPC, NPCIL).



Defense & Paramilitary HAZMAT Squads:

CRPF, CISF airport emergency teams, Indian Coast Guard, and Municipal Fire Rescue Services.



MULTIDIMENSIONAL BENEFITS



SOCIAL & OPERATIONAL SAFETY:

- First responders build reflex-level muscle memory before ever entering a real hot zone.
- Completely eliminates the physical danger of live-agent drills.



ECONOMIC SCALABILITY:

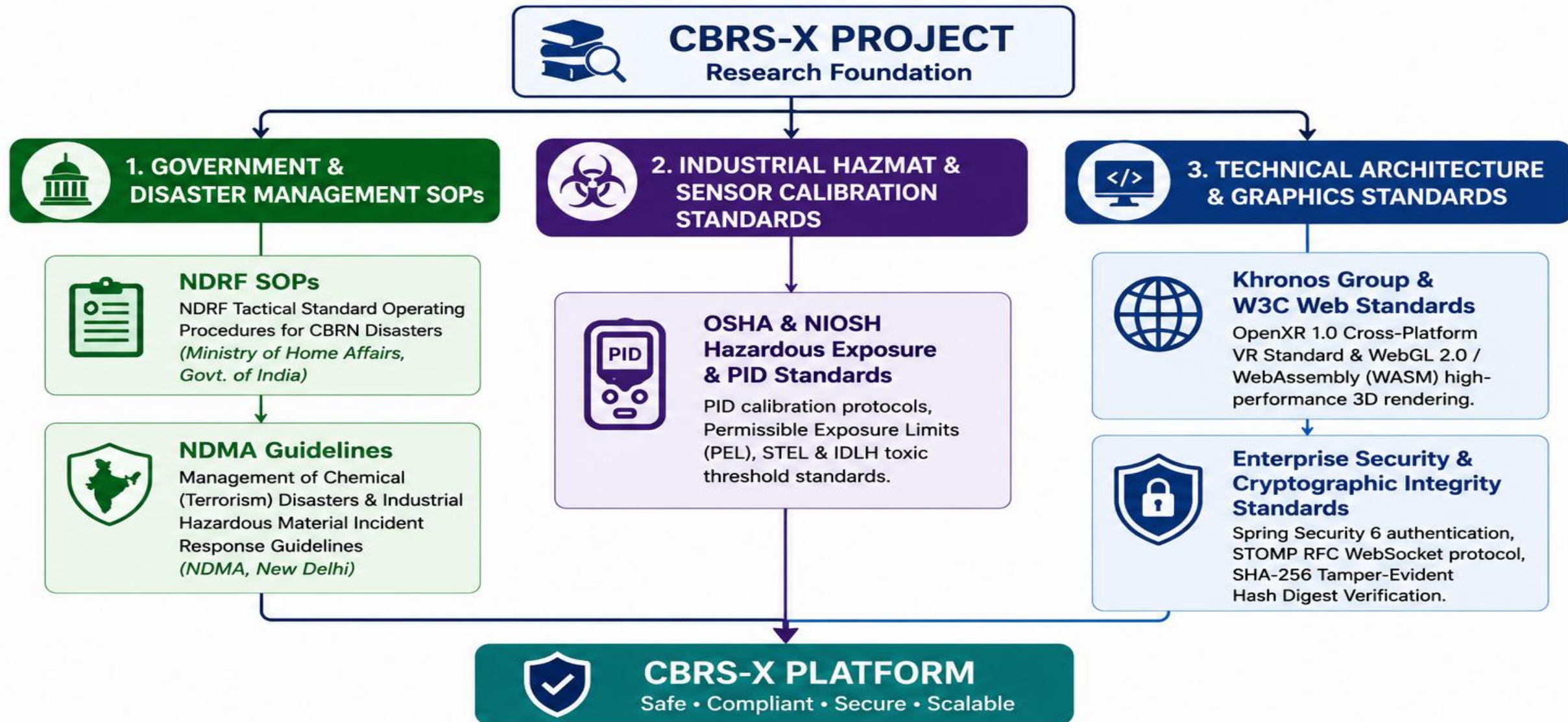
- A single centralized server can train thousands of personnel simultaneously across regional training academies at near-zero marginal cost.



ENVIRONMENTAL SUSTAINABILITY:

- Generates zero chemical runoff and eliminates the disposal of non-biodegradable hazardous material suits into landfills.

RESEARCH BASE, COMPLIANCE & SCIENTIFIC CITATIONS

**COMPLIANCE STATUS:**

Verified against official NDRF tactical response matrices with 100% deterministic test coverage.